

EXPERT REVIEW

The dappled nature of causes of psychiatric illness: replacing the organic–functional/hardware–software dichotomy with empirically based pluralism

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Our tendency to see the world of psychiatric illness in dichotomous and opposing terms has three major sources: the philosophy of Descartes, the state of neuropathology in late nineteenth century Europe (when disorders were divided into those with and without demonstrable pathology and labeled, respectively, organic and functional), and the influential concept of computer functionalism wherein the computer is viewed as a model for the human mind–brain system (brain = hardware, mind = software). These mutually re-enforcing dichotomies, which have had a pernicious influence on our field, make a clear prediction about how ‘difference-makers’ (aka causal risk factors) for psychiatric disorders should be distributed in nature. In particular, are psychiatric disorders like our laptops, which when they dysfunction, can be cleanly divided into those with software versus hardware problems? I propose 11 categories of difference-makers for psychiatric illness from molecular genetics through culture and review their distribution in schizophrenia, major depression and alcohol dependence. In no case do these distributions resemble that predicted by the organic–functional/hardware–software dichotomy. Instead, the causes of psychiatric illness are dappled, distributed widely across multiple categories. We should abandon Cartesian and computer-functionalism-based dichotomies as scientifically inadequate and an impediment to our ability to integrate the diverse information about psychiatric illness our research has produced. Empirically based pluralism provides a rigorous but dappled view of the etiology of psychiatric illness. Critically, it is based not on how we wish the world to be but how the difference-makers for psychiatric illness are in fact distributed.

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Glory be to God for dappled things —
For skies of couple-colour as a brindled cow;
For rose-moles in all stipple upon trout that swim;
Fresh-firecoal chestnut-falls; finches’ wings;
Landscape plotted and pieced — fold, fallow, and plough;
And all trades, their gear and tackle and trim...

Pied Beauty, Gerard Manley Hopkins (1844–1889)¹

The difficulties [in understanding the etiology of insanity] are, in the first place, due to the fact that, as a rule, a number of causal factors work together to induce the resultant insanity.

Richard von Krafft-Ebing (1840–1902),² (p. 136)

Psychiatry has been confused long enough by Descartes’ error. Famously, he postulated that the

human mind/brain system (HMBS) was constituted from two fundamentally different kinds of ‘stuff’: the brain, which was made up of physical ‘stuff’ and occupied space, and the mind, which was made up of thinking ‘stuff’ and was insubstantial. Critically, in the Cartesian worldview, mind and brain became incompatible and opposing ways of thinking about causes of human behavior.

The profound impact of Descartes’ dualistic thinking on psychiatry is best understood through the lens of two subsequent historical developments. The first occurred in the late nineteenth and early twentieth centuries where foundations were laid for the modern disciplines of psychiatry and neurology.^{3,4} Applying the rapidly improving techniques of gross and microscopic neuropathology, some disorders of the HMBS were consistently associated with neuropathological findings and others were not. Although diverse influences were at play, Cartesian philosophy and nineteenth century neuropathology were major contributors to the bargain from which emerged the modern fields of psychiatry and neurology. Neurology dealt with disorders of the HMBS that produced

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consistent neuropathological findings. Psychiatry got what was left. Disorders were either organic (aka brain based) or functional (aka mind based).

In the second major development, in the latter third of the twentieth century, the digital computer came to influence many aspects of modern life. More importantly for our purposes, the computer gave rise to the functionalist view of the HMBS. In functionalism—the dominant philosophical approach to the mind–body problem to this day⁵—the HMBS is understood as an information-processing system directly analogous to a digital computer. The brain is the hardware. The mind is the software constituted by many information-processing programs ‘running’ in the brain. The computer functionalist conceptualization of the HMBS postulates a sharp division between its two parts: hardware and software.

The influences of this dualistic view on psychiatry, derived from a ‘perfect storm’ of Cartesianism, nineteenth century neuropathology and computer functionalism, are legion. Most important has been the functional–organic dichotomy that has dominated major conceptualizations in psychiatry for over a century. An especially pernicious feature of this approach is its ‘either/or’ ness. Disorders are either organic/hardware or functional/software problems.

The effects of the functional–organic dichotomy are insidious. It has been officially dropped from our nosology with DSM-IV,⁶ and most modern psychiatrists and neuroscientists when pressed will deny being dualists, insisting that they are ‘eclectic’ and/or fully recognize that psychiatric disorders are ‘multi-factorial.’ Yet, their actions belie their words. In the ways that we think about patients and their treatments, and the etiology of our disorders a strong tendency remains for us to emphasize either a mentalistic mind-based or a biological brain-based view of illness.⁷

The term ‘mental’ is still in the title of our diagnostic manual and the section of the National Institutes of Health that funds much of our research. Although we might wish that this term would reflect only the psychological nature of the symptoms displayed by our patients (delusions, sadness and panic attacks, which are all experienced subjectively in the first person), inevitably the term ‘mental’ carries with it connotations of etiology—mental and not physical, mind and not brain, functional and not organic. The problem with this approach, which remains embedded in our vocabulary and way of thinking, is that it assumes a discontinuity in the HMBS that does not exist in nature.

Let me now give two vignettes to ‘bring home’ this point—the deep influence of the ‘either/or’ ness of the organic functional divide within our field. First, Gary Greenberg is recounting a contact with his dynamically oriented psychotherapist in which Greenberg is describing to her his severe anxiety in the setting of a depressive episode (Greenberg⁸ p. 19, *italics in the original*).

“Well, what do you think this means?” she asked
“Maybe nothing. I have to say, it felt, I don’t know, *biological*.”

“Biological? You mean, like there are little bugs swimming in your blood or something, making you feel dread?”

“She said this as if it were the most preposterous idea in the world, as if anyone who believed it was either evading the truth or just plain deluded.”

Contrary to much data available then and now, Greenberg’s therapist assumed *a priori* that the explanation of his depressive symptoms was functional—in the realm of the mind. The second vignette is a personal story from earlier in my training, when I was reviewing with a senior biological psychiatry colleague my struggles trying to help a young man with schizophrenia (SZ) to gain insight into the unreality of his frequent referential perceptions. This colleague’s reaction was, ‘I can see wanting to medicate him, but spending a lot of time talking? That is like playing with a computer that is badly broken. Why would you want to do that?’ This senior clinician assumed *a priori* that all SZ symptoms were brain based and, therefore, trying to improve them by cognitive therapy was a waste of time (again, contrary to evidence^{9,10}).

Before proceeding, let me pause to do a bit of needed philosophy. Our experience of the HBMS has an important discontinuity in it. Only we can experience our own consciousness directly and in the first person. However, we have shared experience with others in the third person of behaviors, self-reports, and scientific observations of various aspects of brain structure and function. So from the perspective of knowledge—what philosophers call epistemology—the HMBS has an important first and third person divide. But in terms of what exists—what philosophers call ontology—no such discontinuity exists. The HMBS provides us many levels of analysis with no fundamental difference between psychological and physical phenomenon. Although this is not the place to delve deeply into the metaphysics of the mind–body problem, I here assume non-reductive physicalism, which posits that psychological processes arise in a non-spooky manner from the brain.¹¹ In so doing, these processes have direct causal effects which are not, at least in practical terms, reducible to brain function. So, ontologically, no discontinuity in the HMBS exists in nature.

In a continuation of a line of thought developed in previous essays,^{11,12} I seek to disentangle psychiatry from dualistic thinking through an analysis of the causal factors that research has shown to contribute to psychiatric and substance use disorders. Descartes’ error and computer functionalism assume that psychiatric disorders should naturally sort themselves into two groups: brain/hardware based and mind/software based. But in reality, psychiatric disturbances, all of which arise in the HMBS, reflect a multi-layered ‘dappled’ world on which many causal processes impact and intertwine in their influences.

In reviewing these causal factors, I am looking at how nature has distributed her 'difference-makers' with respect to psychiatric illness. Let me clarify this key concept. Imagine I strike a match and light a candle. The candle's lighting requires a whole range of background factors such as oxygen in the atmosphere, a match that strikes and a candle that lights. But my action was the difference-maker in that if I had not struck the match, despite the presence of all the other background factors, the candle would not be lit. In philosophy, the concept of a difference-maker reflects the counterfactual or 'interventionist' theory of causality^{13,14} and is similar to what epidemiologists would call a risk factor. Being exposed to a difference-maker increases the probability of illness but the difference-maker need not be necessary or sufficient.

This paper has a central thought experiment that generates a clear prediction for the results of psychiatric research. I will show that this prediction is false.

You come to work in the morning and turn on your computer. Instead of the reassuring welcoming screen, you get a garbled mess. In a panic, for you have much work to do before seeing your first patient, you call your IT support person. She arrives quickly and reassuringly gets to work. You chat with her about the problems of differential diagnosis. There are, she says, two main possibilities. Either it is a software problem—perhaps a virus that came with that music video you downloaded last night—or a hardware problem—perhaps your mother board needs to be replaced. "Is it that clear" you ask, "either hardware or software?" "Yes" she replies.

So, if we took all malfunctioning computers, we could, with minimal overlap, divide them into those with software problems and those with hardware problems. The "treatment" approach differs dramatically—that is downloading an anti-viral software program and setting it to work, or taking out your motherboard and shipping it off for a replacement.

Does this story represent a good model for psychiatric illness as Descartes and computer functionalism might wish us to believe? We could live in a universe where psychiatric disorders were like dysfunctional computers where problems arose either from the hardware or the software. But, as I show, we do not.

My argument has a short and a long version. The short version begins with a long list of difference-makers that alter risk for psychiatric and substance use disorders by impacting on the HMBS largely through psychological and social processes. These would include such factors as poor parenting,¹⁵ childhood sexual abuse,¹⁶ stressful life events,¹⁷ severe trauma exposure,¹⁸ coping strategies,¹⁹ social support,²⁰ and exposure to deviant peers.²¹ These risk factors and others like them contribute etiologically to most psychiatric disorders. The next step in the argument is to note that genetic risk factors—which impact on the HMBS through DNA base pair variation influencing protein abundance or structure and eventually brain function—have been shown to be etiologically important in all major psychiatric and

substance use disorders.²² Therefore, with little fuss, we can conclude that for most psychiatric disorders, risk of the illness-related dysfunction of the HMBS results from processes at both psychological and neurobiological levels. Therefore, the idea that psychiatric disorders and patients can, like dysfunctional computers, be cleanly sorted into those with mental=software problems and brain=hardware problems, is false.

Although succinct, this argument is a bit facile. Let me therefore turn to my more ambitious second argument. I seek to implement the concept of a 'causal signature' for psychiatric disorders: a bird's eye view of the distribution of known difference-makers. I will review succinctly and selectively difference-makers for three well-studied psychiatric disorders: SZ (a relatively rare high heritability disorder often seen as epitomizing the more 'organic' end of psychopathology), major depression (MD) (a common disorder with strong environmental risk factors often seen as having an important 'functional' component) and alcohol dependence (AD) (a substance use disorder that demonstrates the special set of difference-makers so associated).

Will the causal signatures resemble those predicted for computers? Will we see all the causal factors clustered either at the hardware/brain end of the continuum or the software/mental end? Or, will we see a much more 'dappled' world with difference-makers sprinkled across many levels?

Causal signatures need a heuristic structure of levels at which causal effects impact on risk for psychiatric and substance use disorders. I assume three super-ordinate categories (biological, psychological and 'higher-order') each with sub-categories that roughly approximate a progression from 'simpler' to more 'complex'. I assume five categories of biological effects: (i) molecular genetic, (ii) molecular/neurochemical neuroscience, (iii) systems neuroscience (including both anatomy and function), (iv) aggregate genetic effects (not yet specified at the molecular level) and (v) miscellaneous biological influences. I assume three categories of psychological effects: (i) neuropsychology, (ii) personality and cognitive/attitudinal patterns and (iii) trauma exposure. Finally, I assume three categories of higher-order effects: (i) social, (ii) political and (iii) cultural. Furthermore, for pragmatic reasons, I initially assume an independence of difference-makers that does not exist in nature. I pick-up this theme below.

After reviewing the distribution of difference-makers for SZ, MD and AD, I assigned a rough score representing the variance in risk that each causal level contributes to the disorder. I forced the totals in the 11 'levels' to sum to 100%. This is a necessarily subjective exercise. My goal is to present a heuristically informative review with no claim that these estimates are exact. Where possible, I reference review articles and meta-analyses. I see this exercise as illustrating an empirically based pluralism.

Schizophrenia

Molecular genetics

Despite tremendous efforts, robustly replicated common genetic variants that impact on schizophrenia have been difficult to identify. The largest effort to date, the Schizophrenia Psychiatric GWAS Consortium, identified seven genome-wide significant findings all of very small effect size (odds ratio ~ 1.1).²³ Other replicated risk variants continue to be reported^{24,25} all of small effect. Copy-number variants that impact on schizophrenia risk have also been replicated and have much larger effect sizes, but are individually rare.²⁶

Molecular neuroscience

Substantial efforts over many decades have gone into developing neurochemical and, more recently, molecular theories of schizophrenia. Almost every major neurotransmitter has been implicated, at one point or another, of playing a critical role in the etiology of schizophrenia, with greatest focus on the dopamine, glutamate and GABA systems.^{27–29} More recent work has combined anatomical, molecular and neurochemical perspectives to develop more specific etiological-anatomic theories (for example, Volk and Lewis³⁰).

Systems neuroscience

Evidence for alterations in brain structure in schizophrenia are both old and very strong³¹ with less consensus on the specific structures involved or the nature of the lesions. Longitudinal studies have now shown evidence for progressive brain changes in both gray and white matter especially in frontal, parietal and temporal areas.³² A number of functional abnormalities have been extensively researched in schizophrenia including abnormalities in event-related potentials and sensory-gating (especially via prepulse inhibition).³³

Aggregate genetic effects

Adoption and twin studies show consistent and strong evidence for aggregate genetic effects in schizophrenia with the best estimates of heritability being quite high ($\sim 80\%$).³⁴ To date, only a very small proportion of this risk has been indexed by known molecular variants. However, recent analysis of GWAS data for schizophrenia suggests that much of this aggregate genetic effect can indeed be explained by common genetic variants of very small effect size.^{23,35}

Other biological risk factors

A wide range of other putative biological risk factors have been examined for schizophrenia and found to be likely difference-makers including obstetric complications,³⁶ season of birth,³⁷ extensive exposure to cannabis,³⁸ and intrauterine viral exposure,³⁷ and famine.³⁷ In general, effect sizes for these risk factors are modest to moderate.

Neuropsychology

A range of neuropsychological deficits have been shown in individuals with schizophrenia, their relatives and unaffected individuals at high risk. Probably the best demonstrated have been attentional abnormalities, but deficits in working memory and a range of executive functions have also been well documented.³³

Personality and cognitive/attitudinal patterns

High risk studies have uncovered a range of traits that are moderately predictive of future risk of schizophrenia including poor social competence and schizotypal features.³⁸

Trauma exposure

While social stressors adversely affect the course of schizophrenia, evidence that they impact on etiology is modest at best.³⁹ The association with various childhood traumas is similarly weak.³⁷

Social

Urbanization is modestly and causally related to risk for schizophrenia.^{37,40} The degree to which this risk is mediated by biological or social processes is currently unclear. Increased risk for schizophrenia is associated with migration, particularly of individuals of African extraction moving to Europe,^{40,41} and at least some of this effect is likely to be causal. Replicated results show that the incidence of schizophrenia in minorities residing in mixed neighborhoods increases as the proportion of minorities in that community decreases.^{38,40} This strongly suggests a social etiologic mechanism.

Political

I was unable to find any compelling evidence that political factors impact on risk for schizophrenia that is not better understood through other levels (for example, famine resulting from war).

Cultural

While cultural factors have been shown to be related to prognosis, compliance and explanatory models for schizophrenia,^{42,43} the best study to date was unable to find ethnic differences in rates of schizophrenia.⁴⁴ Evidence for global variation in incidence of schizophrenia is also weak.³⁸

Major depression

Molecular genetic

Genetic variants that contribute significantly to risk for MD have, to date, been difficult to detect and replicate.^{45–47}

Molecular neuroscience

Many efforts have been made to develop pathophysiological theories for MD based on a number of key neurotransmitters and neuromodulators, for example serotonin, norepinephrine and CRF,^{48,49} and abnormalities in neuroplasticity.⁴⁸

Systems neuroscience

Structural and functional MRI studies have suggested a range of CNS abnormalities that distinguish individuals with MD from controls, including lateral ventricle enlargement and smaller volumes of the basal ganglia, thalamus, hippocampus and orbito-frontal cortex.^{48,50,51}

Aggregate genetic effects

Twin studies consistently show evidence for moderate aggregate genetic effects in MD with the best estimates of heritability of around 40%.⁵²

Other biological risk factors

A number of physiological abnormalities in neuro-endocrine⁵³ and immune function,⁵⁴ and BDNF levels,⁴⁸ have been associated with risk for MD.

Neuropsychology

Research has indicated, with moderate consistency, a range of potential neuropsychological changes in depression, including heightened right-hemisphere interaction, altered emotion processing, attention biases and deficits in working memory.^{55–58}

Personality and cognitive/attitudinal patterns

Several aspects of personality are strongly correlated with risk for MD—especially neuroticism.^{59,60} This association is almost certainly causal.⁶¹ Various dysfunctional cognitions such as hopelessness have been strongly associated with risk for MD^{58,62} and their causal role has been demonstrated by many RCTs of cognitive behavioral therapy.⁶³

Trauma exposure

A range of early environmental risk factors have been well established for MD (for example, poor parenting, sexual abuse),^{64,65} are generalizable across cultures,⁶⁶ and are probably causal.⁶⁷ Stressful life events are strongly associated with risk for MD and^{68,69} much, but not all, of this association is likely causal.⁷⁰

Social

Social factors can impact on risk for MD via levels of unemployment, social disruption and poverty.^{71–73}

Political

I have been unable to find consistent evidence for the impact of political factors on risk for MD that is not mediated through social influences and/or trauma exposure.

Cultural

Cultural factors can shape the expression and help-seeking behavior of those with depressive syndromes.^{74,75}

Alcohol dependence

Molecular genetic

One molecular variant found only in East Asian populations inactivates the aldehyde dehydrogenase

gene and is strongly protective against risk for AD.^{76,77} Variants in the alcohol dehydrogenase gene are more weakly related to risk for AD and are more widely dispersed across human populations.⁷⁶ Functional variants in bitter-taste receptors that reduce the sensitivity to bitter stimuli are associated with risk for AD.^{78,79} Of the range of other replicated molecular variants, the most robust are in the GABA system—in particular in GABA2 subunit genes.⁸⁰ However, the effects of these variants on risk for AD are individually very small with odds ratios, for example, of ~1.15.⁸¹

Molecular neuroscience

Pharmacological studies have suggested that a range of neurochemical systems are involved in the adaptation to heavy alcohol consumption and subsequent addiction including GABA, glutamate, dopamine and opioids.⁸²

Systems neuroscience

A range of studies has suggested that dysfunctional neural systems predispose to AD. One large body of research suggests that individuals at increased risk for AD demonstrate electrophysiological abnormalities manifest both in resting EEGs and in several event-related potential paradigms, particularly a low P300 response.⁸³ Imaging studies suggest that the frontal lobes and their connections with limbic and other cortical regions are compromised in subjects at risk for developing AD.⁸³

Aggregate genetic effects

Twin and adoption studies provide convincing evidence that aggregate genetic risk factors impact strongly on liability to AD with an estimated heritability of 50–60% (for example, Prescott and Kendler⁸⁴; Kendler *et al.*⁸⁵; Heath *et al.*⁸⁶; Pickens *et al.*⁸⁷).

Neuropsychology

Individuals at high risk for AD demonstrate frontal executive deficits,^{88,89} particularly those involving attentional and visuospatial tasks.^{90–92}

Personality and cognitive/attitudinal patterns

Personality traits that reflect either negative emotionality or impulsiveness/novelty seeking predispose to AD.^{93–97} Longitudinal samples beginning in childhood or adolescence (for example, Dubow *et al.*⁹⁶; Englund *et al.*⁹⁷) have shown these traits predict AD suggesting that the associations are probably causal.

Two sets of attitudes influence risk for AD. The first of these is alcohol expectancies^{98–101} manipulation of which in some, but not all, studies impacts on subsequent risk for AD.¹⁰² The second are 'reasons for drinking' which also impact on risk for heavy alcohol consumption and AD (for example, Abbey *et al.*¹⁰³; Farber *et al.*¹⁰⁴).

Trauma exposure

Several early childhood adversities have been associated with risk for AD including parental loss,^{105,106}

poor parent-child relationships^{107,108} and childhood sexual abuse.¹⁶ Twin based twin and twin family methods have examined the impact of parental loss¹⁰⁵ and childhood sexual abuse^{67,109} and suggest that most of the observed association is causal.

Social

Risk for AD is robustly predicted by social factors such as peer substance use, drug availability and social class.^{93,110} Confirming these observations are ecological studies of college populations which show very strong correlations between heavy-drinking and drinking-related problems, and the density of near-by alcohol outlets (for example, Weitzman *et al.*¹¹¹) and twin studies which consistently show that adolescent alcohol use is strongly influenced by shared environmental effects.^{112,113} Furthermore, a Cochrane review shows that college students receiving feedback via the web or computer about normative levels of alcohol consumption have significantly reduced alcohol-related problems and binge drinking.¹¹⁴

Political

Two detailed meta-analyses^{115,116} conclude that the pricing of alcohol and its availability impact substantially on the risk for AD and related adverse effects of alcohol use and that a good deal of this relationship is likely causal. More specific factors that have been studied and shown to impact on risk include size of alcohol beverages,¹¹⁷ hours in which pubs and bars are open,¹¹⁸ and the geographical distribution and density of retail outlets where alcohol can be purchased.^{119,120}

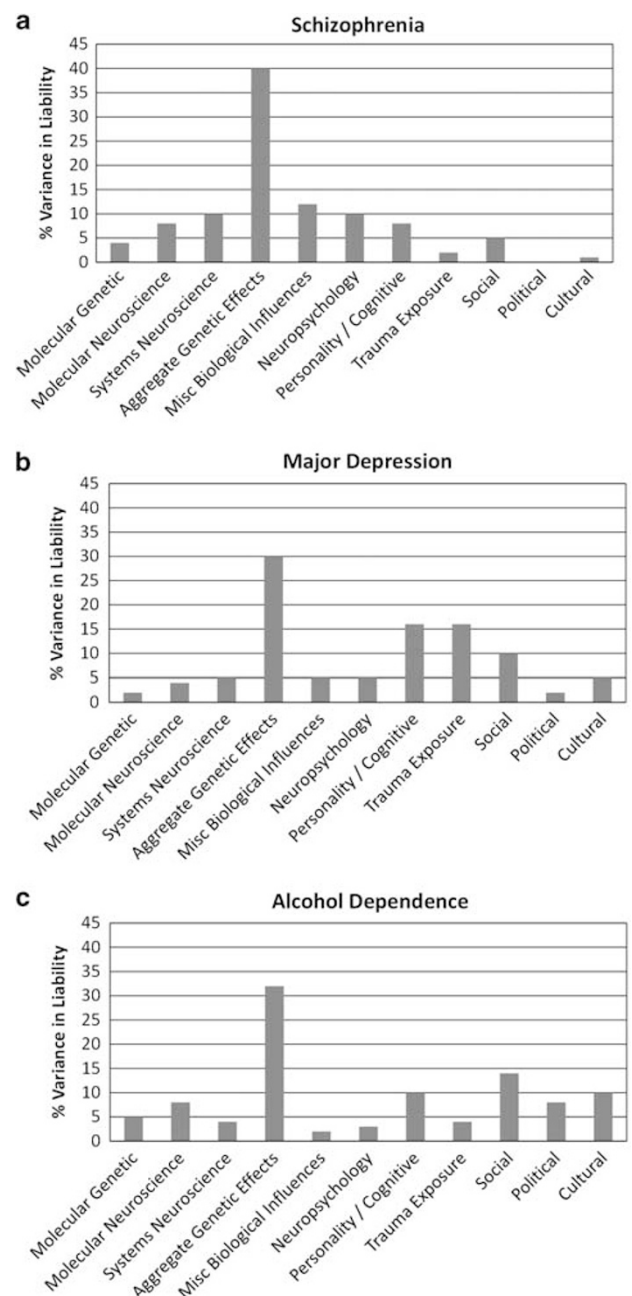
Cultural

A wide range of strong cultural factors have been shown on rates of AD including religious beliefs,¹¹⁰ the preferred form of ethanol (for example, beer, wine, spirits),¹²¹ the acceptability of public drunkenness,¹²² and the appropriateness of drinking by men versus women (which strongly influences gender-specific risk for AD).¹²³ In migrant and native populations, rates of AD often rise with the breakdown of traditional cultural practices and beliefs.^{94,124}

Causal signatures

This exercise in empirically based pluralism is summarized in Figures 1a–c, which depict causal

Figure 1 (a) A causal signature for schizophrenia. The 11 'bins' of difference-makers (aka risk factors) are organized along the x axis from the 'lowest' or most 'basic' (molecular genetic) to the highest or most complex (cultural). The y axis reflects the % of variance in liability to illness accounted for by difference-makers in each of these bins. These estimates arose from a detailed review of the relevant literature but inevitably involve some subjective judgment. (b) A causal signature for major depression. The 11 'bins' of difference-makers (aka risk factors) are organized along the x axis from the 'lowest' or most 'basic' (molecular genetic) to the highest or most complex (cultural). The y axis reflects the % of variance in liability to illness accounted for by difference-makers in each of these bins. These estimates arose from a detailed review of the relevant literature but inevitably involve some subjective judgment. (c) A causal signature for alcohol dependence (broadly defined to include serious alcohol related problems). The 11 'bins' of difference-makers (aka risk factors) are organized along the x axis from the 'lowest' or most 'basic' (molecular genetic) to the highest or most complex (cultural). The y axis reflects the % of variance in liability to illness accounted for by difference-makers in each of these bins. These estimates arose from a detailed review of the relevant literature but inevitably involve some subjective judgment.



signatures for SZ, MD and AD. Three major conclusions can be drawn. First, difference-makers for these three psychiatric disorders are dappled, spreading over biological, psychological and high-order domains. There is no evidence, as predicted by computer functionalism, that psychiatric disorders can be cleanly divided into those that reflect hardware versus software problems.

Second, for all three disorders, aggregate genetic effects make the largest single contribution, largely because these influences have been well studied and quantified.

Third, the specific patterns of these three disorders differ meaningfully. SZ has more difference-makers in the biological arena, MD in the psychological arena and AD in the higher-order domains. However, each of these three disorders has important difference-makers across all domains. Although the specific features of the causal signatures will differ across disorders, the spread of difference-makers across these domains is likely a general feature of psychiatric disorders. The HMBS is, by its nature, sensitive to a wide range of causal processes.

The tidy pictures of Figures 1a–c imply that the impact of each level of difference-maker is independent of each other. However, as illustrated elsewhere for MD⁶⁴ and AD,¹²⁵ the truth is far more complex. The influences of high-level difference-makers often flow through lower level processes. For example, the impact of stressful life events on risk for depression¹⁷ is likely mediated by changes at more basic psychological and neurobiological levels. Lower level difference-makers can also impact on risk for illness through higher-level processes. Genes can alter risk for AD via influencing self-selection of deviant peers¹²⁵ and on depression via their effect on personality.⁶¹ The causal paths of the difference-makers start at one level but often flow through others on their way to influencing disease risk. Furthermore, the effects of individual binds of difference-makers are often non-additive. Genetic factors can moderate the impact of stressful events on risk for MD¹²⁶ while key features of the social environment can alter the impact of genetic effects on alcohol intake.¹²⁷ But this complication only strengthens my argument. Not only are difference-makers for psychiatric disorders dappled across the traditional mind–brain or hardware–software divide, but difference-makers sitting on either side of these putative divisions often mediate and moderate each other's effects.

Concerns and comments

I tried to test the predictions of the two major modern versions of dualistic thinking within psychiatry: the organic–functional and hardware–software dichotomies. Few readers of the modern psychiatric literature will be surprised at the findings. If the results of our science are accepted, for the three archetypal disorders examined, difference-makers are distributed across the biological, psychological and

social–cultural domains, and these levels are actively inter-twined with each other in etiologic pathways. These results support my main contention. Our field needs to disentangle itself from the still influential ghost of Descartes and adopt, in its stead, an empirically informed pluralism.

The conceptual framework here advocated has been focused at the level of disorders, not individuals. Perhaps the research reviewed here and the subsequent causal signatures reflect a false complexity. Perhaps the computer-functionalism model still works for psychiatric illnesses, but at the level of individuals and not disorders. So, for example, cases of MD could be cleanly divided into ‘psychogenic’ forms—that have mind-based software problems—and ‘biological’ forms—that reflect brain-based hardware problems. However, both clinical and research experience indicate that this is not the case. Psychogenic forms of MD (aka ‘reactive’ or ‘situational’) do not differ from ‘non-psychogenic’ forms in their level of familial/genetic risk for MD.^{128,129} In a large cohort of depressed subjects, we found no relationship between the level of adversity associated with onset and the familial risk of MD,¹³⁰ as would be predicted if cases could be sorted into psychogenic and genetic/biological forms. In empirically derived multifactorial etiological models for MD, genetic and environmental risk factors are positively and developmentally inter-twined, not, as predicted by the computer-functionalism model, negatively correlated.^{64,65} A similar picture is seen for AD.¹²⁵ Unlike dysfunctional computers, our patients do not simply sort themselves into those that have hardware/brain versus software/mind problems.

This picture of difference-makers widely distributed across multiple levels is not unique to psychiatric disorders. For some biomedical syndromes, difference-makers exist on only one level (for example, Mendelian disorders and molecular genetic variation). But for other syndromes (for example, coronary heart disease, type 2 diabetes), the picture would likely resemble that seen for psychiatric illness, including a significant number of difference-makers from the social and cultural bins.

How does the empirically based pluralism differ from the biopsychosocial model of Engel?¹³¹ The crucial difference is that it is not *a priori*—driven by a theoretical commitment to pluralism—but rather driven directly by what our research reveals. The biopsychosocial model—while making us feel good about our open-mindedness—offers us no critical guidance.¹³² As McHugh and Slavney cogently comment ‘... the biopsychosocial model is heuristically sterile’ (McHugh and Slavney¹³³ p. 288). In letting all flowers bloom, it provides no focus. By contrast, research has increasingly told us where to look, but it is surely not in only one place. For our various disorders, some bins of difference-makers will be quite full and others much emptier. Furthermore, our research is providing cautious insights into etiologic pathways. Ultimately the question, of course, is not

only which difference-makers are involved but also how they join together to cause the disease. In moving away from the antiquated dichotomies of Descartes and computer functionalism, we need to be open minded about where nature has put the difference-makers, but hard nosed in following what our research tells us rather than our prior expectations.

Empirically based pluralism offers one further hope—that it can guide psychiatry beyond our history of rancor and inter-denominational warfare. It requires only the acceptance of a common metric—the quality of research evidence. Otherwise, all perspectives on psychiatric illness should be given equal treatment.

Empirically based pluralism can help place in perspective claims like ‘panic disorder is a brain disorder’ that have become increasingly common in the last several decades. Such a claim is only meaningful in the context of the organic–functional dichotomy. It might be seen as true in the sense that brain- and biology-related difference-makers contribute to risk for panic disorder and false in the sense that it claims that panic is a ‘hardware only’ kind of disorder. Although such claims reflect an appropriate need for medical and social legitimacy for psychiatric disorders, they are neither entirely true nor terribly helpful in the context of empirically based pluralism. The view that only by being ‘biological’ does a psychiatric disorder become ‘real’ is a symptom of the Cartesian error.

The model of empirically based pluralism is vulnerable to reductionist attack that might take the following form: you claim to find difference-makers in psychological and social constructs. But how do they work? Surely, the real effects are occurring in brain even if the causal chain starts with mental or social processes. Your empirically based pluralism all collapses to neurobiology.

I have two responses: one pragmatic and the other theoretical. The pragmatic approach is

While you can worry about your theoretical reductions, where the world gives us causal traction is critical. If I can treat MD with psychotherapy¹³⁴ or show that suicide rates decline substantially after major social stressors,¹³⁵ then these are real causal effects. Sure, they are likely mediated by neurobiological processes but that does not make these psychological or social risk factors epiphenomenal. After all, everything that goes on in brain is a result of chemical and ultimately subatomic processes. Should we then conclude that the only “real” causes of psychiatric illness are at the level of quarks?

Theoretically, I would argue that the level of psychological is critical for many forms of psychiatric illness. This can be no better demonstrated than with the question of meaning which I illustrate with a single research example. In our study of the effects of CSA, rates of MD were increased over threefold in women exposed to the most severe forms of sexual abuse: attempted or completed intercourse.⁶⁷

However, we also asked the women whether they told anyone about the abuse and if so what happened? If they told someone and got a negative response, risk increased another 50%. However, if they told someone and the abuse stopped, the risk for lifetime MD returned almost to control levels.¹³⁶ What was pathogenic about the abuse was not the physical acts, but the meaning of that event for the young girl. Did she feel abandoned and unprotected by those who should be caring for her? Once that sense of care and protection was re-established the future excess risk for MD nearly disappeared. I am not suggesting that feelings of abandonment and isolation do not have a neurobiology. The point is that the feeling of abandonment is here the key difference-maker.

The messy results found in our analysis of causal signatures are not consistent with simple models of consilience of our world¹³⁷ in which sociology reduces easily to psychology, psychology to biology, biology to chemistry, and chemistry to physics. Our results reflect a more complex and dappled worldview well captured by the philosophers John Dupré¹³⁸ and Nancy Cartwright.¹³⁹

Why are we so different from computers? Does this reflect the differences between an engineered machine and an evolved organism? What does it take to develop a HMBS that functions well in those areas, like affect regulation and reality testing, which are disturbed in psychiatric illness? Recall that evolution is a tinkering gradual process. Natural selection will not produce or sustain genetic programs for experiences that are typically available in the environment. Consider our prolonged childhoods with sustained parental care, our tight social groups with well worked-out status relationships, the long-term pair bonding strongly associated with good adjustment and the symbolic logic related to language development. An engineered machine can be designed to operate at two distinct levels (that is, hardware and software). However, for a complex evolved organism like us, the product of eons of evolutionary tinkering, multiple inter-twined genetic and environmental influences will be needed for healthy functioning. For each of these influences there will be, in multi-level complex causal webs, difference-makers that can push development down pathological pathways.

I also need to comment on the functional–organic concept as it came down to use from late nineteenth century Germany. It is not sensible to assume that—whatever our technology—if we can ‘see it’ in the brain then it is ‘organic’ and if it is not, it is ‘functional’. The approach is fallacious and only makes sense if you assume a Cartesian worldview and you are looking for where to ‘split’ the HMBS. If, by contrast, you accept that mind does not exist in a spirit form free of physical instantiation in brain, it does not make sense to assign a fundamentally different status to phenomena which produced changes in (1) gross brain structure (for example, reducing size of some key structure of 20%) and so could be seen by the naked eye in a postmortem study

or by routine structural scans versus (2) subtle neuronal structure (for example, increased boutons on a subclass of dendritic spines in certain cell types) that could only be seen by electron microscopy versus (3) physiology function only with no detectable anatomic changes (for example, alterations in key protein concentrations).

Nothing in the essay is inconsistent with the fact that HMBS can be vulnerable to strong biological insults, ranging from Mendelian genetic disorders to gunshot wounds, which, for all practical purposes, act independent of psychological and higher-order functions.

The results of the empirically based pluralistic analysis of the causes of SZ, MD and AD reinforce the conclusions from a prior essay that the commonly expressed wish to develop an etiologically based nosology for psychiatric disorders is deeply problematic.¹⁴⁰ Psychiatric disorders are a result of multiple etiological processes impacting on many different levels and often further intertwined by mediational and moderational interactions between levels. It is not possible *a priori* to identify one privileged level that can unambiguously be used as the basis for developing a nosologic system.

My call for an empirically based pluralism does not reflect pessimism about the future of research in the etiology of psychiatric disorders. Surely, they are stunningly complex. But having overly simplified views of them, often ideologically driven, has only hampered our field. Following methods of decomposition and reassembly, progress has been made in the scientific understanding of very complex systems.^{12,141,142} Having a realistic view of the causal landscapes of psychiatric disorders can only help.

Summary

This essay began with a brief review of the origins of the dualistic thinking that continues to plague our field in Descartes' philosophy, nineteenth century neuropathology and computer functionalism. I posed, as a central question, whether the HMBS becomes disordered the way our laptops do—typically sorting clearly into hardware and software problems. I proposed the concept of difference-makers and causal signatures, and then briefly reviewed these patterns for SZ, MD and AD. Inconsistent with the predictions of dualistic thinking, for all of these disorders, difference-makers are distributed across the biological, psychological and social-cultural spheres. Our deeply entrenched dualistic thinking is a conceptual impediment to our ability to integrate the diverse information about psychiatric illness our research has produced. We need to finally disentangle ourselves from the ghost of Descartes and adopt, in his stead, an empirically based pluralism.

Conflict of interest

The author declares no conflict of interest.

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